

1. Administrative & Study Identification

**Full Study Title:** Study of Efficacy and Safety of Eltrombopag in Lower-risk MDS Patients With Platelet Transfusion Dependence  
**Official Title:** A Randomized, Double-blind, Placebo-controlled, Japan Local Phase II Clinical Study Comparing Eltrombopag Monotherapy Versus Placebo in Adult Lower-risk Myelodysplastic Syndromes (MDS) Patients With Platelet Transfusion Dependence  
**Short Title/Acronym:** Eltrombopag in LR-MDS **Protocol Number:** CETB115L11201 **NCT Number:** NCT04797000 **Posting ID:** 456764293526330337 **Sponsor/Company Name:** Novartis / Novartis Pharmaceuticals **Investigational Product:** Eltrombopag (Trade names: Promacta, Alvaiz) / Placebo (12.5 mg & 25 mg oral tablets)  
**Phase of Development:** Phase II (Phase requirement: Trial must be in PHASE2) **Trial Status:** ACTIVE\_NOT\_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To demonstrate the superiority of eltrombopag versus placebo in terms of the proportion of participants who achieve platelet transfusion independence at Week 24. **Secondary Objectives:** To evaluate the time to platelet transfusion independence, the duration of platelet transfusion independence, the percentage of participants with platelet transfusion independence, relapse after hematological improvement (HI), time to progression to Acute Myeloid Leukemia (AML) (defined as  $\geq 20\%$  blasts at Week 24, Year 1, and Year 2), and overall survival (OS).

**2.2 Background & Rationale** To address the gap in the treatment of severe thrombocytopenia in adult Japanese patients with lower-risk myelodysplastic syndromes (LR-MDS). The management of LR-MDS with persistent cytopenias remains suboptimal. Alternative therapeutic options are needed because platelet transfusions offer only short-term efficacy, and many patients do not respond to hypomethylating agents. Eltrombopag, an oral thrombopoietin receptor agonist (TPO-RA), is being evaluated to improve platelet counts and transfusion independence in this population.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 ratio **Status Name:** ACTIVE\_NOT\_RECRUITING

**3.2 Planned Enrollment & Duration Target** **\$N\$:** 36 patients **Number of Sites:** 20 locations (Japan) **Estimated Study Start:** 25-May-2021 (Actual) **Estimated Primary Completion:** 26-Apr-2025 (Actual) **Estimated Study Completion:** 09-Dec-2026

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age 20 to 100 years. 2. Patients diagnosed with MDS according to the WHO classification revised 4th edition by investigator assessment with one of the following prognostic risk categories, based on the International Prognostic Scoring System (IPSS-R): very low (0-1.5), low (2-3), or intermediate risks (3.5-4.5). \* All following criteria for prognostic variables per IPSS-R should be met: \* Bone marrow blast  $\leq 5\%$  (per both investigator's assessment and central review). \* Cytogenetic very good, good or intermediate risk corresponding to IPSS-R. 3. Platelet transfusion dependence at baseline, defined as receiving platelet transfusion regularly with a frequency of 2 or more times within 4 weeks prior to randomization. Platelet transfusion should be performed for a patient with platelet counts  $\leq 20 \times 10^9/L$ , or with hemorrhagic symptoms and platelet counts  $\leq 30 \times 10^9/L$ . 4. Refractory, intolerant to, or ineligible for MDS treatments. 5. Eastern Cooperative Oncology Group (ECOG) Performance status (PS) 0 or 1.

**4.2 Exclusion Criteria** 1. Patients with a history of prior administration of eltrombopag, romiplostim, or other TPO-RAs. 2. Therapy-related MDS per WHO classification revised 4th edition. 3. MDS/myeloproliferative neoplasms including chronic myelomonocytic leukemia (CMML) per the WHO classification revised 4th edition. 4. MDS with excess blasts (EB) per WHO classification revised 4th edition. 5. Known history of IPSS-R high or very high-risk MDS. 6. Currently receiving treatments for MDS (e.g., hypomethylating agents [HMA], cyclosporine A [CsA], or lenalidomide). *Note: Supportive treatment with ESAs or G-CSF may be allowed under specific conditions (see 5.2).* 7. Patients scheduled for hematopoietic stem cell transplantation. 8. Bone marrow fibrosis that leads to an inability to aspirate adequate bone marrow sample. 9. Known thrombophilic risk factors (except in cases where potential benefits of participating in the study outweighed potential risks of thromboembolic events (TEE), as determined by the investigator).

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen** **Drug Name:** Eltrombopag (ATC Code: B02BX05) **Dosage/Frequency:** Eltrombopag or Placebo 12.5 mg & 25 mg tablets taken once per day (QD). **Route of Administration:** Oral **Duration of Treatment:** Long-term follow-up through Week 24, Year 1, and Year 2.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard supportive care for thrombocytopenia and platelet transfusions as needed. Supportive treatment with erythropoiesis-stimulating agents (ESAs) in anemic patients or granulocyte-colony stimulating factor (G-CSF) in patients with severe neutropenia and recurrent infections is allowed if at stable dosage for 3 months prior to screening and continued at the same dosing/schedule until the optimal dose of eltrombopag has been established. **Prohibited:** Concurrent disease-modifying therapies for MDS (e.g., HMA, cyclosporine A, or lenalidomide).

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                                                   |
|--------------|---------------|-------------------------------------------------------------------------------------------------------------|
| Screening    | Day -14 to 0  | Informed Consent, Medical History, Baseline platelet counts                                                 |
| Baseline     | Day 0         | Randomization 1:1, First Dose of Eltrombopag/Placebo                                                        |
| Interim      | Week 24       | Primary efficacy assessment for transfusion independence, Safety Labs, Central/Local blast count assessment |
| End of Study | Year 1 & 2    | Assessment for progression to AML ( $\geq 20\%$ blasts), Overall Survival follow-up                         |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Percentage of participants who achieve platelet transfusion independence at Week 24. Platelet transfusion independence is defined as the absence of platelet transfusion for at least 8 weeks. Platelet count should be higher than the baseline count. **Measurement Method:** Laboratory analysis (platelet counts) and transfusion record reviews.

**7.2 Secondary & Exploratory Endpoints**

- \* Time to platelet transfusion independence.
- \* Duration of platelet transfusion independence (from randomization to the first day of a platelet transfusion-free period lasting at least 8 weeks).
- \* Percentage of participants with platelet transfusion independence.
- \* Relapse after hematological improvement (HI) and transfusion independence.
- \* Time to progression to AML (defined as  $\geq 20\%$  blasts at Week 24, Year 1, and Year 2) or death.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Routine collection via clinical assessments (e.g., assessing bleeding events, hepatotoxicity, thromboembolic events [TEE], and reticulin fibrosis typical of TPO-RAs). **Serious Adverse Events (SAEs):** Routine reporting timeline, typically 24 hours. **Data Monitoring Committee (DMC):** Internal/External safety monitoring mechanisms managed by Novartis.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy superiority over placebo. Safety Set for all AE/SAE evaluations. **Sample Size Justification:**  $N=36$  powered to demonstrate the superiority of eltrombopag versus placebo for achieving platelet transfusion independence at Week 24. **Handling of Missing Data:** Standard handling for missing data in hematology trials; patients lost to follow-up are typically evaluated as non-responders.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular monitoring of clinical sites to ensure protocol adherence and safety. **Audit Trail:** Both investigator assessment and central review are utilized per modified IWG criteria and WHO classification (for blast percentages and hematologic improvement).

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## 11. Study Identifiers & Governance

**Primary ID:** CETB115L11201 **Posting ID:** 456764293526330337 **Registry Number:** NCT04797000 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Eltrombopag in LR-MDS **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled, Japan Local Phase II Clinical Study Comparing Eltrombopag Monotherapy Versus Placebo in Adult Lower-risk Myelodysplastic Syndromes (MDS) Patients With Platelet Transfusion Dependence

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## 12. Clinical Framework (MDS Specific)

**Primary Condition / Disease Area:** Myelodysplastic Syndromes  
**Name / Preferred Name:** Myelodysplastic Syndrome / MYELODYSPLASTIC SYNDROME **Preferred UMLS Name:** Low-Risk Myelodysplastic Syndromes  
**Semantic Type:** Neoplastic Process **Definition:** A clonal hematopoietic disorder characterized by dysplasia and ineffective hematopoiesis in one or more of the hematopoietic cell lines. The dysplasia may be accompanied by an increase in myeloblasts, but the number is less than 20%, which, according to the WHO guidelines, is the requisite threshold for the diagnosis of acute myeloid leukemia. It may occur de novo or as a result of exposure to alkylating agents and/or radiotherapy. (WHO, 2001) **Clinical Codes & Classifications:** \* **MeSH Code:** D009190 \* **HPO Code:** HP: 0002863 \* **SNOMED ID:** 109995007 \* **SNOMED Hierarchy:** \* 269475001 | Malignant neoplasm of lymphoid, hematopoietic and/or related tissue (disorder) \* 425333006 | Myeloproliferative disorder (disorder) \* 414823004 | Neoplasm affecting hematopoietic

structure (disorder) \* 414824005 | Neoplasm of bone marrow (disorder) \* 14016003 | Bone marrow structure (body structure) \* 128623006 | Myelodysplastic neoplasm (morphologic abnormality) \* 64572001 | Disease (disorder) **Medical Methodology:** Thrombopoietin receptor agonist (TPO-RA) **Optimization Target:** Platelet transfusion independence and hematological improvement (HI).

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### 13. Study Procedures & Methodology

**Management Pathway:** Best supportive care combined with Eltrombopag or Placebo. **Education Protocol (HCP):** Protocol training on TPO-RA safety, blast count monitoring, and central review criteria. **Education Protocol (Patient):** Informed consent and instructions on oral QD medication compliance. **Quality Audit Mechanism:** Central review of modified IWG criteria.

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### 14. Observation & Follow-Up Schedule

**Total Duration:** Up to Year 2 / study completion. **Onsite Visit Cadence:** Regular clinical visits through Week 24, Year 1, and Year 2. **Remote Monitoring Frequency:** As specified by Novartis protocol parameters. **Checklist Review Interval:** Regular laboratory assessments for platelet counts and blasts.

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### 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | ____      | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | ____      | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | ____      | [DD-MMM-YYYY] |     |      |     |     |